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Anomaly detection using surprisals

Abstract

TBC

Keywords: Extreme value theory; Generalized Pareto Distribution; Outlier detection; Tail bounds.

1 Introduction

Anomaly detection methods are often rather ad hoc and lack a solid theoretical foundation, or they make strong distributional assumptions that may not hold in practice. We aim to address this by defining an anomaly as a point of low probability, and proposing a general framework for anomaly detection by estimating the probability of each observation, or one more anomalous. Our approach allows for a unified treatment of anomaly detection that can be applied to any anomaly detection problem, provided a probability distribution can be defined on the space of observations. It allows for univariate and multivariate settings, and can be applied to both unconditional and conditional distributions. Although it requires a specified probability distribution, it is robust to substantial mis-specification of that distribution.

The basic idea is to calculate the surprisal (equal to the negative log density) of each observation, and then estimate the tail probability from the surprisal distribution. This allows us to estimate the probability of observing a surprisal value as extreme or more extreme than a given observation, which can then be used to flag anomalies. The surprisal approach to anomaly detection provides a principled way to control the false positive rate when flagging anomalies, by setting a threshold on the estimated tail probabilities.

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ be iid observations from a probability distribution F defined on an appropriate sample space, and let f be the generalized probability density function of F. That is, f is defined using Dirac delta functions at jump points of F, so that it represents the density function if F is continuous, the probability mass function if F is discrete, and a combination of both if F is a mixture of continuous and discrete components.

The *surprisal* of observation $\boldsymbol{y}_i$ is defined as (Stone 2022)

$$s_i = -\log f(\boldsymbol{y}_i).$$

While these are better known as "log scores" in statistics (Good 1952), in this context we prefer the term "surprisal" (coined by Tribus 1961), as it captures the idea that observations with low density are more "surprising" (and therefore potentially anomalous) than those with high density. A large value of s_i indicates that $\boldsymbol{y}_i$ is an unlikely value, and so is a potential anomaly. On the other hand, typical values will have low surprisals. In information theory, the average surprisal is known as the entropy of a random variable (Cover & Thomas 2006; Stone 2022) and provides a measure of uncertainty. If f is obtained from a parametric model, then the sum of the surprisals is equal to minus the log likelihood of the data.

Let $G(s) = \Pr\{S \leq s\}$ be the distribution function of the surprisal values, where $S = -\log f(\boldsymbol{Y})$ and $\boldsymbol{Y} \sim F$. Then, the probability of observing a value with surprisal at least as large as s is given by $1 - G(s)$, which leads to a natural way to assign an anomaly score to each observation $\boldsymbol{y}_i$:

$$p_i = 1 - G(s_i) = 1 - G(-\log f(\boldsymbol{y}_i)).$$

We define anomalies as observations with $p_i < \alpha$, giving a false positive rate of α if all observations are from F. This is preferable to directly computing the probabilities from the tail of F, as it allows for anomalies to occur anywhere in the distribution of F, not just in its tails. For example, anomalies may lie between modes in a multimodal distribution, or in otherwise low-density regions of the distribution F. It is also preferable to a distance-based approach to anomaly detection (e.g., Wilkinson 2018), as it naturally handles skewed and heavy-tailed distributions.

The surprisal approach converts the problem of working with F, a potentially multivariate distribution (or even one on a manifold or some non-Euclidean space), to working with a univariate distribution G defined on $\mathbb{R}$. We can write

$$G(s) = \Pr\{-\log f(\boldsymbol{Y}) \leq s\} = \Pr\{f(\boldsymbol{Y}) \geq e^{-s}\}.$$

This makes clear that the surprisal probability is related to the highest density regions of the distribution (Hyndman 1996). In fact, every observation with $p_i < \alpha$ lies outside the $100(1-\alpha)\%$ highest density region of F.

We will usually not know F, and so we need to estimate it from data or from theory. Such estimates are particularly fragile or uncertain in areas of low density, which is precisely where we are looking for anomalies. Further, we are seeking a method that does not make strong distributional assumptions.

Our approach begins with an assumed F, which may be estimated from data, or obtained from theory. This distribution is used to compute the surprisal values $\{s_1, \dots, s_n\}$. Then, we estimate the tail

probabilities $\{p_1, \dots, p_n\}$ from the surprisal values, using one of three approaches.

(1) Using the assumed distribution F to compute the tail probabilities, i.e., $p_i = \int I\{f(\boldsymbol{y}) \leq e^{-s_i}\} d\boldsymbol{y}$, where I is the indicator function;
(2) Using the empirical distribution function of the observed surprisals $\{s_1, \dots, s_n\}$, so that p_i is the proportion of surprisals at least as large as s_i;
(3) Fitting a Generalized Pareto Distribution (GPD) to the largest surprisal values, and using this to estimate the tail probabilities.

Obviously approach (1) makes the strong assumption that F is an accurate representation of the data. Approaches (2) and (3) allow for considerable mis-specification of the distribution used to compute the surprisals. Approach (2) requires only that the ordering of the surprisals in the tail is similar to what would have been obtained under the true distribution, while approach (3) requires that extreme value theory applies to the surprisal distribution.

We illustrate the three approaches with several examples, including both simulated and real data.

> **Note**
>
> Outline of paper TBC

2 Empirical surprisal probabilities

Under approach (2) above, we estimate p_i as the proportion of surprisals larger than s_i. For large n, this provides a quick and convenient estimate of p_i that is accurate even when the assumed distribution F is not.

> **Important**
>
> What can we say about the quality of the approximation?
> If the surprisal ranks under the true and estimated F distributions are equal, this is obviously ok.
> That should be true in the tails of any distribution — i.e., when y_i is greater than the largest mode of f, or less than the smallest mode of f.
> Can we say anything else?

Using this approach, the anomaly boundary for a threshold of α corresponds to the $100(1-\alpha)\%$ highest density region computed using the quantile density approach of Hyndman (1996).

3 Extreme value theory for the surprisal distribution

Our third approach uses extreme value theory applied to the surprisal values. We will show that under weak conditions on the tail of the surprisal distribution, the maximum surprisal converges to one of the three types of extreme value distributions. This justifies fitting a Generalized Pareto Distribution (GPD) to the largest surprisal values, and using it to estimate tail probabilities.

An important precursor to this work is Kandanaarachchi & Hyndman (2022) who proposed estimating F by a leave-one-out kernel density estimate, and then fitting a Generalized Pareto Distribution (GPD) to the most extreme surprisal values. The underlying idea is that even if F is not particularly well-estimated, the tail of the surprisal distribution should still be well-approximated by a GPD. Hyndman, Kandanaarachchi & Turner (2025) provides some theoretical justification for this approach in the context of kernel density estimation.

Here we extend that idea to allow F to be any well-behaved distribution, including conditional distributions. Specifically, we fit a GPD to the largest β of $\{s_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters, (μ, σ, ξ). We calculate the probability of each observation or one more extreme, $p_i = \beta\left[1 - P(s_i \mid \hat{\mu}, \hat{\sigma}, \hat{\xi})\right]$, where P is the distribution function of the GPD. Observations not contained in the largest β of $\{s_i\}_{i=1}^n$ are assigned $p_i = \beta$. To justify this approach, we seek conditions on the surprisal distribution of $S = -\log f(\boldsymbol{Y})$, where $\boldsymbol{Y} \sim F$, similar to the Fisher-Tippett-Gnedenko (FTG) theorem (Coles 2001).

We consider three possible conditions on the surprisal distribution of $S = -\log f(\boldsymbol{Y})$ where f is the density of $\boldsymbol{Y}$. These conditions describe increasingly heavy tails:

1. where S is sub-Gaussian (e.g., when f is bounded above and below);
2. where S is sub-exponential (e.g., when f is Gaussian); and
3. where S has a polynomial moment (e.g., when f is Student-t).

We state these as three alternative assumptions.

Assumption 3.1 (Sub-Gaussian)**.** The random variable $S = -\log f(Y) \in \mathbb{R}$, and satisfies, for all $\lambda \in \mathbb{R}$, and some $\nu > 0$,

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\}.$$

Assumption 3.2 (Sub-exponential)**.** The random variable S is sub-exponential with parameters ν and b, i.e.,

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\} \quad \text{for all} \quad |\lambda| < 1/b.$$

Assumption 3.3 (Polynomial)**.** The random variable $|S|$ has polynomial moments of order $\alpha \geq 1$; i.e., $\mathbb{E}[|S|^p] \leq C^p$ for some $C > 0$ such that $C^p - 1 > 0$.

Under each of Assumptions 3.1–3.3, we seek a result on the limiting distribution of the maximum of the surprisals $M_n = \max\{s_1, \dots, s_n\}$. If we can show that this distribution falls within the class of extreme value distributions, then we can fit a Generalized Pareto Distribution (GPD) to the largest surprisal values (Coles 2001).

3.1 S has sub-Gaussian tails

Let's first assume that the surprisals are well-behaved and have a sub-Gaussian tail. Assumption 3.1 will be satisfied in cases where the density f is bounded both above and below, such as with mass functions on a finite set. However, it may not be satisfied if the log density has no lower bound.

For example, consider the truncated normal distribution on $[a, b]$, where $Y \sim \mathcal{TN}(a, b, \mu, \sigma^2)$ and $-\infty < a < b < \infty$. Then

$$S = -\log(2\pi\sigma^2)/2 - \frac{1}{2\sigma^2}(Y-\mu)^2 - \log\{\Phi(b) - \Phi(a)\},$$

where Φ is the standard normal distribution function. Since $Y \in [a, b]$, S is bounded for all $\sigma > 0$ and hence S satisfies Assumption 3.1.

Under Assumption 3.1, we have the following result on the maximum of the surprisals.

Lemma 3.1. *Suppose $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \dots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If $S = -\log f(Y) \in \mathbb{R}$, and satisfies, for all $\lambda \in \mathbb{R}$, and some $\nu > 0$,*

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\},$$

, then, for $s \in \mathbb{R}_+$,

$$\Pr\{M_n - \mathbb{E}[S] \geq s\} \leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.$$

Proof. For any $\lambda > 0$,

$$\begin{aligned}\Pr\{S \geq s\} = \Pr\left\{e^{\lambda(S-\mathbb{E}[S])} \geq e^{\lambda(s-\mathbb{E}[S])}\right\} &\leq e^{-\lambda(s-\mu)}\mathbb{E}e^{\lambda(S-\mu)} \\ &\leq e^{-\lambda(s-\mu)}e^{\lambda^2\nu^2/2}.\end{aligned}$$

Now, we can maximise the RHS of the above bound in λ, which yields the optimal value $\lambda =$

$(s-\mu)/\{M^2\}$. Plugging this in we have the bound

$$\Pr\{S \geq s\} \leq e^{-\frac{1}{2}(s-\mu)^2/\nu^2}.$$

The stated result now follows by noting that

$$\begin{aligned}\Pr(M_n - \mathbb{E}[S] \geq s) = 1 - \Pr\{M_n \leq s\} &= 1 - \Pr\{S \leq s\}^n \\ &= 1 - (1 - \Pr\{S \geq s\})^n \\ &\leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.\end{aligned}$$

□

Corollary 3.1. *If Assumption 3.1 is satisfied, then, for all $s > 0$,*

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(2n)}\right) \leq e^{-s}.$$

Proof. This follows directly from the union bound:

$$\begin{aligned}\Pr\left(|M_n| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}\right) &\leq n \Pr\left\{|S| \geq \sqrt{2}\nu^2 s + \sqrt{2\nu^2 \log(n)}\right\} \\ &\leq 2ne^{-[\sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}]^2/(2\nu^2)} \\ &= 2ne^{-\frac{1}{2\nu^2}(2s\nu^2) - \frac{2\nu^2 \log(2n)}{2\nu^2} - \frac{\sqrt{2\nu^2 s}\sqrt{2\nu^2 \log(2n)}}{2\nu^2}} \\ &\leq e^{-\sqrt{\log(2n)}s - s} \\ &\leq e^{-s}\end{aligned}$$

The first result follows from the iid assumption on S and the union bound, while the second from the assumption that S is sub-Gaussian. The remaining inequalities follow from reorganising terms. □

Corollary 3.1 is useful as it clarifies that if the surprisal is sub-Gaussian, then the maximal surprisal will follow a reversed Weibull distribution, and thus is within the class of extreme value distributions.

3.2 S has an exponential moment

In cases where the underlying density has full support on $\mathbb{R}$ and the log density cannot be bounded from below, then the sub-Gaussianity condition in Assumption 3.1 may be violated, as the following result illustrates.

Lemma 3.2. *If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $S = -\log f(Y)$ is sub-exponential. In particular,*

$$\mathbb{E}e^{\lambda(S-1/2)+(\lambda/2)\log(2\pi\sigma^2)} \le e^{\lambda^2/2+(\lambda/2)\log(2\pi\sigma^2)}, \quad \forall|\lambda| < 1/2.$$

Proof. Firstly, note that

$$S = -\log f(Y) = \frac{1}{2}\log(2\pi\sigma^2) + \frac{(Y-\mu)^2}{2\sigma^2}.$$

and that

$$\mu_S = \mathbb{E}[S] = -\int f(y)\log f(y)\mathrm{d}y = \frac{1}{2}\{1+\log(2\pi\sigma^2)\}.$$

Define

$$X = \frac{1}{2}\frac{(Y-\mu)^2}{\sigma^2} - \frac{1}{2},$$

then $\mathbb{E}[X] = 0$. Now, note that

$$\mathbb{E}e^{\frac{\lambda}{2}(S-1)} = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{1}{\sqrt{2\pi\sigma^2}}\int e^{\frac{\lambda}{2}\left[\frac{(y-\mu)^2}{\sigma^2}-1\right]}e^{-\frac{(y-\mu)^2}{2\sigma^2}}\mathrm{d}y.$$

Consider the change of variables $z = (y-\mu)/\sigma$ so that we have

$$\mathbb{E}e^{\frac{\lambda}{2}(S-1)} = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{1}{\sqrt{2\pi}}\int e^{\frac{\lambda}{2}[z^2-1]}e^{-z^2/2}\mathrm{d}y = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{e^{-2(\lambda/2)}}{\sqrt{1-2(\lambda/2)}}$$

for all $\lambda < 1$. Further, it can be verified that

$$\frac{e^{-\lambda/2}}{\sqrt{1-2(\lambda/2)}} \le e^{\lambda^2/2}, \quad \forall|\lambda| < 1/2.$$

Hence, when we have $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $S = -\log f(Y)$ is sub-exponential with parameter $\nu = 1$ and $b = 2$. □

Therefore, to be able to apply extreme value theory to the surprisal values, we need a version of Lemma 3.1 that uses sub-exponentiality.

Lemma 3.3. *Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 3.2 is satisfied, then, for $s \in \mathbb{R}_+$,*

$$\Pr(M_n - \mathbb{E}[S] \ge s) \le 1 - \begin{cases} (1-e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \le s \le \nu^2/b \\ (1-e^{-s/2b})^n & \text{if } s > \nu^2/b \end{cases}.$$

Proof. The result follows a similar argument as for Lemma 3.1, up until we maximize the tail bound.

In particular, for any $\lambda > 0$,

$$\begin{aligned}\Pr\{S - \mathbb{E}[S] \geq s\} = \Pr\left\{e^{\lambda(S-\mathbb{E}[S])} \geq e^{\lambda s}\right\} &\leq e^{-\lambda s}\mathbb{E}e^{\lambda(S-\mathbb{E}[S])} \\ &\leq e^{-\lambda s}e^{\lambda^2\nu^2/2}.\end{aligned}$$

Now, we must consider a constrained optimization to ensure the right hand side exists. In particular, we know from the proof of Lemma 3.1 that the unconstrained optimum is $\lambda = s/\nu^2$, which corresponds to the constrained optimum on the set $0 \leq t < \nu^2/b$. However, when $s \geq \nu^2/b$, the maximum is achieved at $\lambda = 1/b$, and putting this together yields

$$\Pr\{S - \mathbb{E}[S] \geq s\} \leq \begin{cases} e^{-\frac{1}{2}s^2/\nu^2} & \text{if } 0 \leq s \leq \nu^2/b; \\ e^{-s/2b} & \text{if } s > \nu^2/b. \end{cases}$$

The stated result again follows by noting that

$$\begin{aligned}\Pr\{M_n - \mathbb{E}[S] \geq s\} &= 1 - \Pr\{M_n \leq s\} \\ &= 1 - \Pr\{S \leq s\}^n \\ &\leq 1 - \begin{cases} (1 - e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b; \\ (1 - e^{-s/2b})^n & \text{if } s > \nu^2/b. \end{cases}\end{aligned}$$

□

Results similar to Lemma 3.3 also hold for $\Pr\{M_n - \mathbb{E}[S] \leq s\}$, and for $\Pr\{|M_n - \mathbb{E}[S]| \geq s\}$ with an additional factor of 2, i.e.,

$$\Pr\{|M_n - \mathbb{E}[S]| \geq s\} \leq 1 - \begin{cases} (1 - 2e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b \\ (1 - 2e^{-s/2b})^n & \text{if } s > \nu^2/b. \end{cases}$$

Corollary 3.2. *Under Assumption 3.2, for $s > \nu^2/b$, as $n \to \infty$*

$$\Pr(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)) \leq 1 - e^{-e^{-s}} + o(1/n).$$

Proof. Apply Assumption 3.2 to see that

$$\begin{aligned}\Pr(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)) &\leq 1 - \left\{1 - 2e^{-[2bs+2b\log(2n)]/2b}\right\}^n \\ &= 1 - \left(1 - e^{-s}/n\right)^n\end{aligned}$$

$$= 1 - e^{e^{-s}} - \left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right].$$

The result now follows by noting that, for all $s > 0$,

$$\left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right] = o(1) \leq 0.$$

□

Corollary 3.2 is useful as it shows that if the surprisal has a sub-exponential tail, then the resulting maximum surprisal tail probability converges to that of an extreme value distribution. In particular, we have that Corollary 3.2 implies

$$\sup_{s:s>\nu^2/b} \left|\Pr\left\{|M_n - \mu| \geq \left\lceil 2bs + 2b\log(2n)\right\rceil\right\} - \left(1 - e^{e^{-s}}\right)\right| = o(1).$$

To see that this implies that $M_n - \mathbb{E}[S]$, appropriately standardized, converges to an extreme value distribution, we first recall that

$$\Pr\{X_n > x\} - \Pr\{X > x\} = 1 - \Pr\{X_n \leq x\} - [1 - \Pr\{X \leq x\}] = \Pr\{X \leq x\} - \Pr\{X_n \leq x\}.$$

Hence,

$$\sup_{x\in\mathscr{X}} |\Pr\{X_n \leq x\} - \Pr\{X \leq x\}| = \sup_{x\in\mathscr{X}} |\Pr\{X_n > x\} - \Pr\{X > x\}|,$$

so that when X_n and X are both continuous we see that Corollary 3.2 implies that the CDF of $(M_n - \mathbb{E}[S] - 2b\log(2n))/2b$ converges to that of the Type-1 extreme value distribution (i.e., Gumbel distribution); i.e., using Corollary 3.2 and the above we have that

$$\sup_{s:s>\nu^2/b} \left|\Pr\left(|M_n - \mu| \leq \lceil 2bs + 2b\log(2n)\rceil\right\} - e^{e^{-s}}\right| = o(1).$$

This is one notion of convergence in distribution.

3.3 S has a polynomial moment

Finally, we consider the case where the surprisal has a polynomial moment (e.g., Student-t random variables), and derive a corresponding result to Lemma 3.1.

Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 3.3 is satisfied, then, for $s > 0$ and such that $s > C > 0$,

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq s\right) \leq 1 - (1 - C^p/s^p)^n.$$

Hence,

$$\sup_{s:s>C} \left| \Pr\left(|M_n - \mathbb{E}[S]| \le (s \cdot n^{1/p})\right) - e^{-s^{-p}} \right| = o(1).$$

Proof. By Markov's inequality,

$$\Pr\{|S - \mathbb{E}[S]| \ge s\} \le \frac{\mathbb{E}|S|^p}{s^p} \le \frac{C^p}{s^p}.$$

To derive the first stated result, again note that

$$\begin{aligned}
\Pr\{|M_n - \mathbb{E}[S]| \ge s\} &= 1 - \Pr\{|M_n - \mathbb{E}[S]| \le s\} \\
&= 1 - \Pr\{|S - \mathbb{E}[S]| \le s\}^n \\
&= 1 - (1 - \Pr\{|S - \mathbb{E}[S]| \ge s\})^n \\
&\le 1 - (1 - C^p/s^p)^n.
\end{aligned}$$

To derive the second stated result, apply the first result with $t = sn^{1/p}C > 1$, which clearly holds for some n large enough, to see that

$$\Pr\{|M_n - \mathbb{E}[S]| \ge t\} \le 1 - \left(1 - \frac{C^p}{(Csn^{1/p})^p}\right)^n = 1 - \left(1 - s^{-p}/n\right)^n.$$

Now, recall that, for $x \ge 0$,

$$\lim_{n\to\infty} (1 - x/n)^n = e^{-x}.$$

Since $x \mapsto e^{-x}$ is bounded for all $x \ge 0$, it follows that

$$\sup_{s:s>C} \left| \Pr\{|M_n - \mathbb{E}[S]| \ge t\} - e^{-s^{-p}} \right| = o(1).$$

□

3.4 Three types theorem

Placing the three lemmas together gives the following "three types" Theorem.

Theorem 3.1. *Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ be an iid sequence from F, and let $S = -\log f(\boldsymbol{Y})$, where $Y \sim \boldsymbol{F}$. Further, let $M_n = \max\{s_1, \ldots, s_n\}$, where $s_i = -\log f(y_i)$.*

1. *If the surprisal satisfies Assumption 3.1, then*

$$\sup_{s:s>0} \left| \Pr\left\{ \left| M_n - \mathbb{E}[S] \right| \ge \sqrt{2v^2 s} + \sqrt{2v^2 \log(2n)} \right\} - e^{-s} \right| = o(1).$$

2. *If the surprisal satisfies Assumption 3.2, then*

$$\sup_{s:s>1/b} \left| \Pr\left\{ \left| M_n - \mathbb{E}[S] \right| \geq (2b)s + (2b)\log(2n) \right\} - e^{-e^{-s}} \right| = o(1).$$

3. *If the surprisal satisfies Assumption 3.3, then*

$$\sup_{s:s>C} \left| \Pr\left\{ \left| M_n - \mathbb{E}[S] \right| \geq (Csn^{1/p}) \right\} - e^{-s^{-p}} \right| = o(1).$$

Theorem 3.1 gives a corresponding 'three types' theorem for the behavior of surprisals and has a simple interpretation. If the surprisal has Gaussian like tails, then the maximum surprisal is a reversed Weibull; if it only has an exponential moment, then the maximum surprisal will be Gumbel; if the maximum surprisal only has a polynomial moment, then it is Fréchet. As each of these distributions is a member of the generalized extreme value family, then the upper tail of the distribution can be approximated by a Generalized Pareto Distribution (Coles 2001, Theorem 4.1).

4 Experiments

4.1 Standard univariate normal values

Although we need to assume some density function f when computing surprisals, we want the resulting tail probabilities to be robust to a mis-specified distribution. Suppose we generate 1000 observations from a N(0,1) distribution, but use a $t(4)$ distribution to compute the surprisal values. The empirical tail probabilities, and those obtained using the GPD approach, should still give reasonable estimates of the tail probabilities, despite the incorrect distribution being used to compute the surprisals.

To study the effect of the assumed distribution, we compute the surprisal values using both the correct Normal density and an incorrect $t(4)$ density. Then, for each set of surprisal values, we estimate the surprisal tail probabilities in three ways: (a) using the same distribution used to compute the surprisal values; (b) using empirical tail probabilities; and (c) using a GPD fitted to the largest 100 surprisal values.

The various estimates are displayed in the left panel of Figure 1 for observations greater than 2.5, showing how both empirical tail probabilities and the GPD estimates are still accurate, regardless of which distribution was used to compute the surprisals. The solid lines show the results for surprisals computed using the true N(0,1) distribution, while the dashed line shows the results for surprisals computed using the mis-specified $t(4)$ distribution. The estimates computed directly from $t(4)$ ("Assumed distribution") are very inaccurate, but all other estimates lie close to the truth.

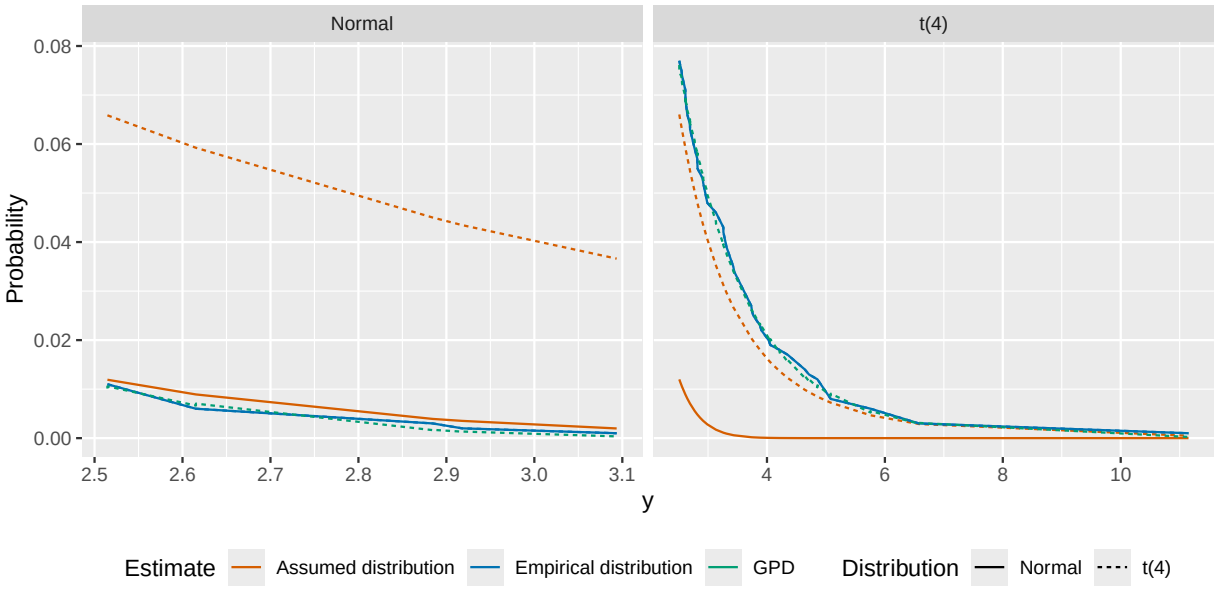


Figure 1: *Left: Observations are computed from a N(0,1) distribution. Right: Observations are computed from a $t(4)$ distribution. Distribution indicates which distribution was used to compute the surprisal values. Estimate shows how the tail probabilities were computed. This demonstrates that the surprisal probabilities computed using the empirical distribution or under the GPD are still relatively accurate, even when the wrong distribution is used to compute the surprisal values.*

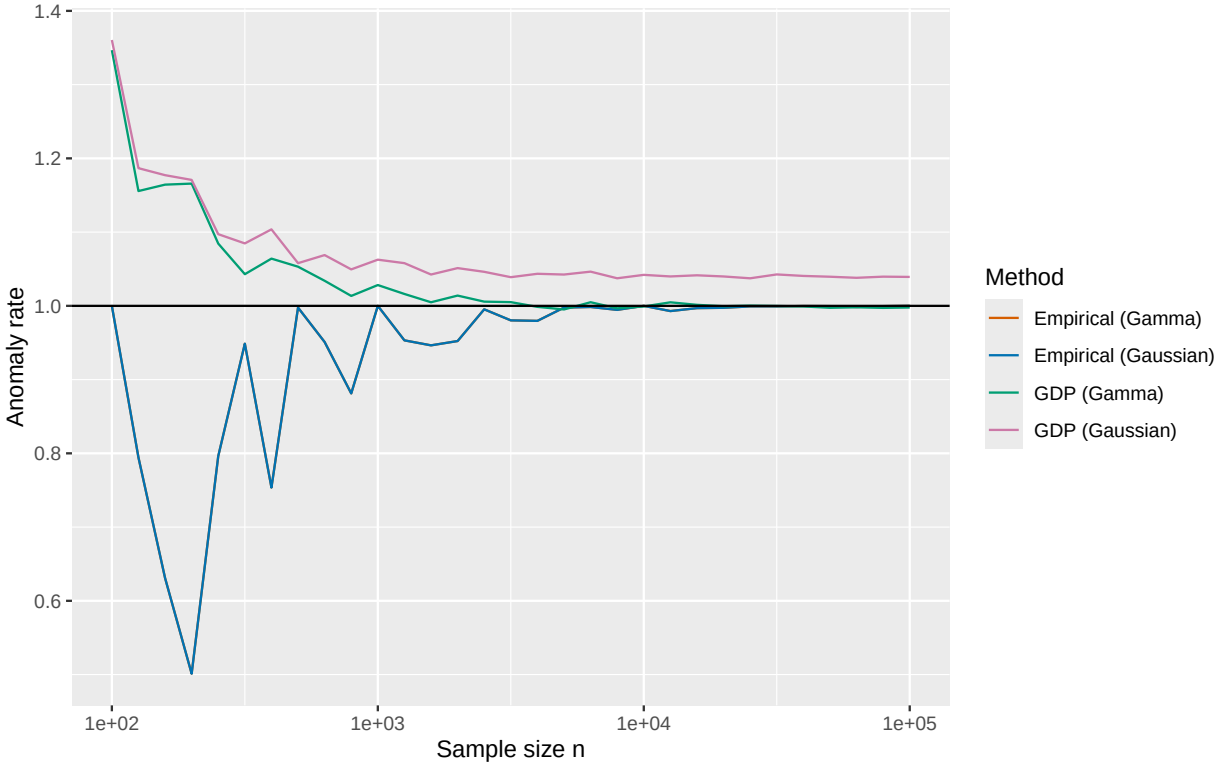
In summary, even when we don't know the distribution of the data, we can use some incorrect assumed distribution to compute the surprisals, and then apply either a GPD or the empirical distribution to obtain good estimates of the surprisal probabilities.

The converse situation is shown in the right panel of Figure 1, where the data are generated from a $t(4)$ distribution, and the surprisal values are computed using both the correct $t(4)$ density and an incorrect N(0,1) density. Again, the empirical tail probabilities, and the GPD estimates, are relatively accurate, even when the wrong distribution is used to compute the surprisal values.

4.2 Bivariate gamma values

The quality of the approximation used depends on the amount of data available, so in this next experiment we compare the tail probabilities of the different approaches across sample sizes. Let $\boldsymbol{Y} = (Y_{i,1}, Y_{i,2})'$ where $Y_{i,j}$ are independent Gamma(2,2) random variables. We generate n observations from this bivariate distribution, and explore the quality of the probability estimates as n increases from 100 to 100,000. The surprisal values are computed using the correct distribution, as well as the incorrect bivariate normal distribution with mean $(1,1)'$ and covariance matrix $0.5\boldsymbol{I}$. For each set of surprisal values, we estimate the tail probabilities in two ways: (a) using empirical tail probabilities; and (b) using a GPD fitted to the largest 10% of surprisal values. We repeat this experiment 1000 times for each sample size. The anomaly rate shown is the number of anomalies detected when

$\alpha = 0.01$ for different sample sizes, averaged over 1000 replications. The false anomaly rate under the true distribution is, of course, 0.01.



5 Applications

- US COVID death rates: https://otexts.com/weird/09-timeseries.html#multivariate-time-series
- https://otexts.com/weird/06-density.html#gam-surprisals

6 Discussion

Software and reproducibility

All results presented here can be reproduced using the code available at https://github.com/robjhyndman/Surprisal_Theory. The analysis was conducted using R version 4.5.1 (R Core Team 2025), with the following R packages: weird (Hyndman 2024), tsibble (Wang et al. 2025), targets (Landau 2025, 2021), ggplot2 (Wickham et al. 2025; Wickham 2016), and other tidyverse (Wickham et al. 2019) packages.

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